

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – SHORT ANSWER TEST

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 – Short Answer Test
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	Sasikumar Megha	Reg. No.:	192471038
Section / Batch:	3 rd CS & E	Date:	04/09/2026

Code of Conduct

I, Sasikumar Megha (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sasikumar Megha

Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Annexure A – Short Answer Test

1. Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Now perform the following operations: 1) Invert the elements in the stack, 2) Pop (), 3) Pop (), 3) Pop Construct AVL tree for the following elements. Mention the
2. Given a collection of elements 10,20,30,40,50,60,70,80. The Key element to be search 70. Compare how searching is performed in Linear Search and Binary Search. Identify the efficient search method. Justify.
3. What is the postfix and prefix forms of the following expression? $A+B*(C-D)/(P-R)$
4. Construct AVL tree for the following elements. Mention the Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10
5. Construct a heap by inserting 10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45 into a binary heap.
6. Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
7. Perform the merge sort for the following inputs. Also, show the result for each step of iteration. 40,20,70,14,60,61,97,30.
8. How the following data set can be sorted by quick sort. Show each step-in sorting. 78,23,31,88,43,55,67,55.
9. Perform Open Hashing & Linear Probing for the given set of 0,1,81, 4,64,25,16,36, 9 and 49.
10. Consider the following graph & construct BFS.



Graph with six nodes

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with minor gaps	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with minor errors	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with minor gaps	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & Completion	10	Completes all questions accurately within the allotted time	Completes most questions on time	Requires additional time	Unable to complete the test

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: <u>21/5/26</u>	Date: _____	Date: _____

Co 5- AT-3

Basikumar Megha

192471038

CSA0305

Stack operations

Given stack size = 5

Position : 0 1 2 3 4
 22 55 33 66 88

Assume Position 4 is the Top.

Step 1 : Invert the stack

88 66 33 55 22

Top = 22

Step 2 : POP()

88 66 33 55

Top = 55

step 3 : POP()

88 66 33

Top = 33

Final Stack : 88 66 33

2) Linear Search Vs Binarch Search.

10, 20, 30, 40, 50, 60, 70, 80

Key = 70

Linear search checks element one by one:

10 $\rightarrow$ 20 $\rightarrow$ 30 $\rightarrow$ 40 $\rightarrow$ 50 $\rightarrow$ 60 $\rightarrow$ 70

Number of comparisons = 7.

Binary Search

$\rightarrow$ The array is already sorted

Middle element : 40

70 > 40. So search right

50 60 70 80

Middle element : 60

70 > 60 search right

70 80

$\rightarrow$ No. of comparisons ≈ 3

* Binarch search is the efficient method because its time complexity is $O(\log n)$ compared with $O(n)$ for linear search.

3) Prefix and Postfix

$$A + B * (C - D) / (P - R)$$

step 1: Identify operators

$$A + [B * (C - D)] / (P - R)$$

Postfix:

for (C - D): CD -

for B * (C - D): BCD - *

for (P - R): PR -

$$\therefore ABCD - * PR - / +$$

Prefix:

$$+ A / * B - CD - PR$$

4) AVL Tree for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

step 1:

1

step 2:

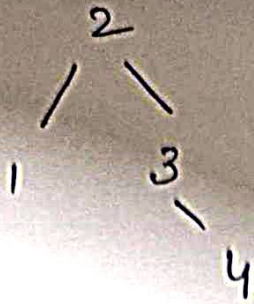
1
└ 2

step 3:

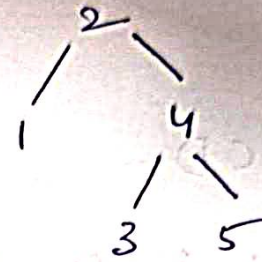
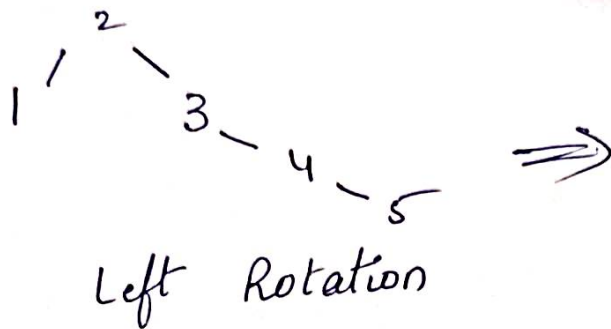
1
└ 2
 └ 3

Left Rotation

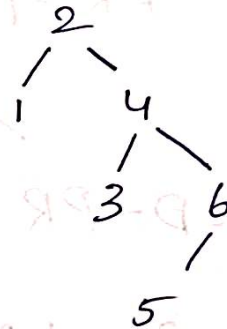
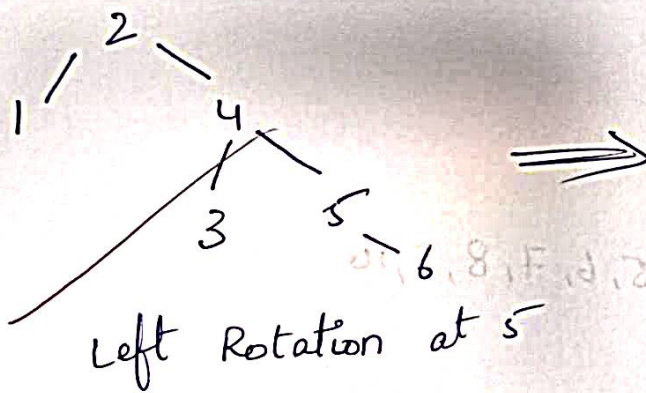
2
└ 1
 └ 3



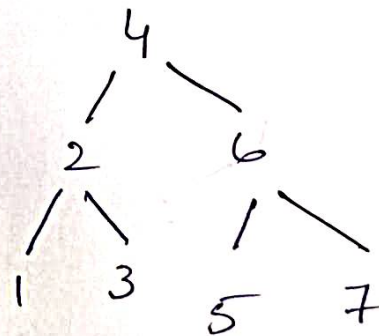
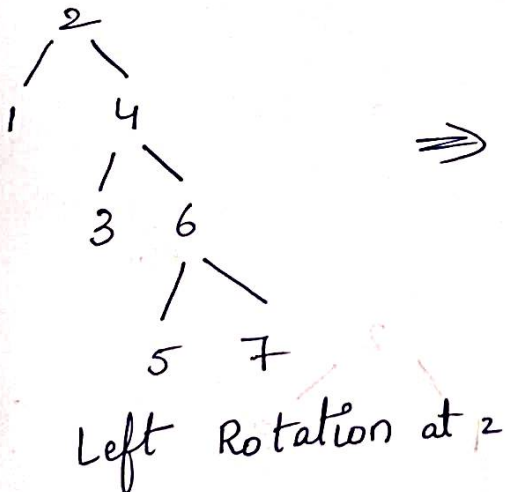
step 5:



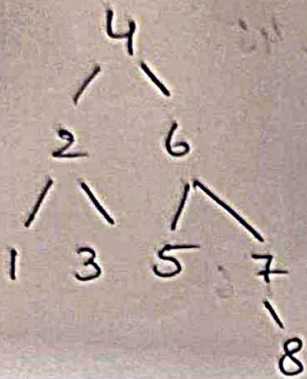
step 6:



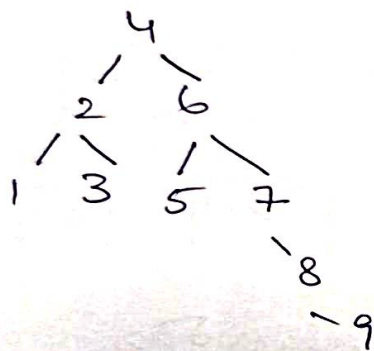
step 7:



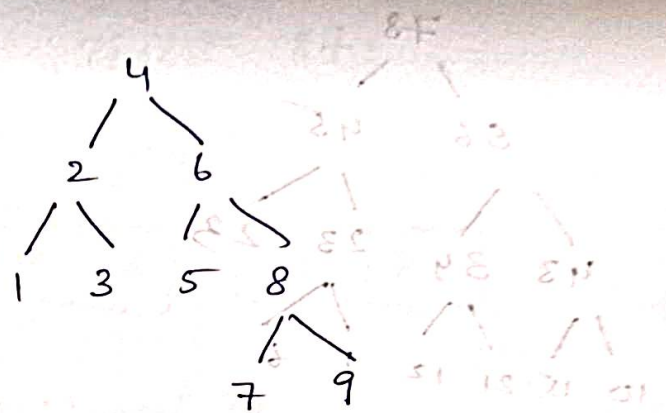
step 8 :



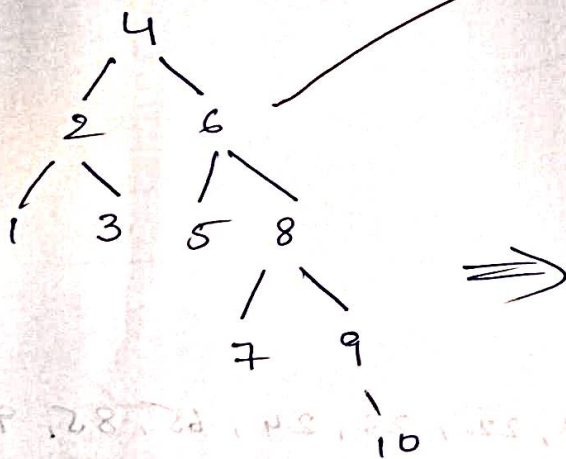
step 9 :



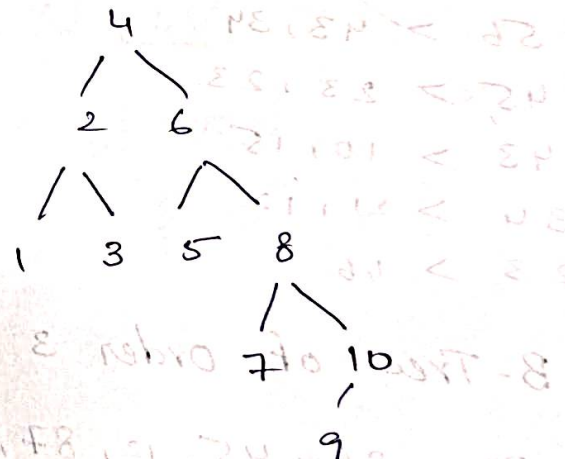
Left Rotation at 7



step 10 :



Left rotation at 7



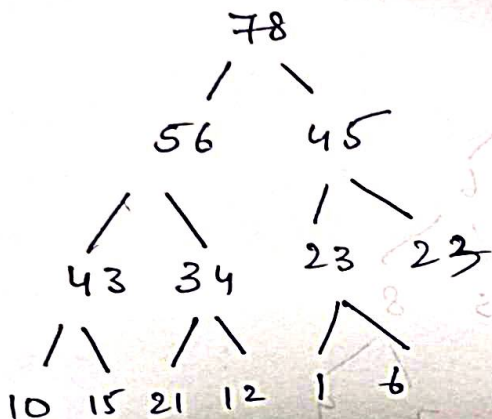
final AVL Tree

5) Binary Heap

10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45

final max Heap

78 56 45 43 34 23 23 10 15 21 12 1 6



Heap Property.

$$78 > 56, 45$$

$$56 > 43, 34$$

$$45 > 23, 23$$

$$43 > 10, 15$$

$$34 > 21, 12$$

$$23 > 1, 6$$

6. B-Tree of order 3

34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95

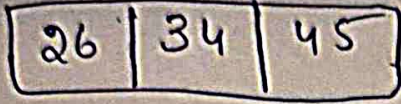
Max children = 3

minimum Keys = 2

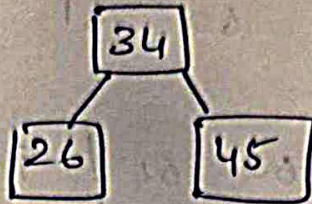
34



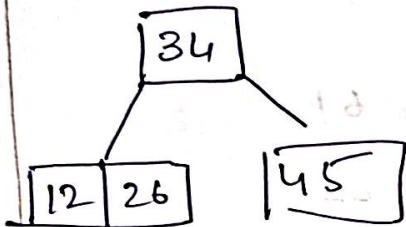
Insert 45:



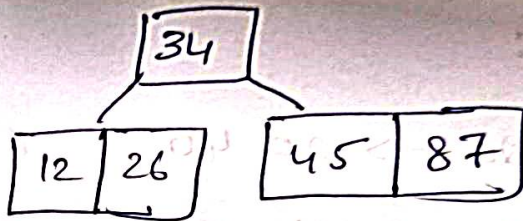
⇒



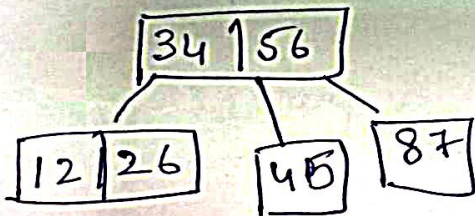
Insert 12



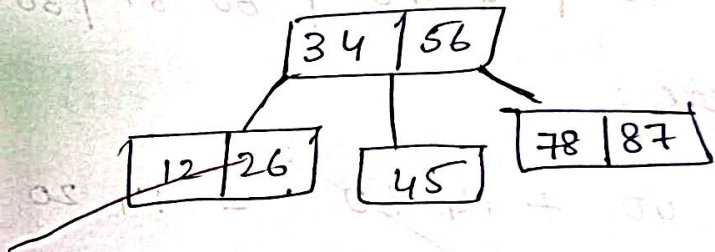
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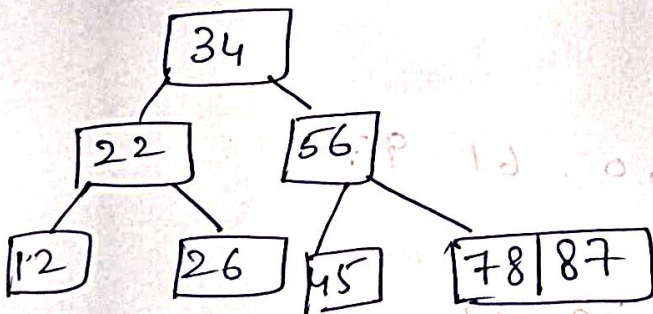
Insert 56



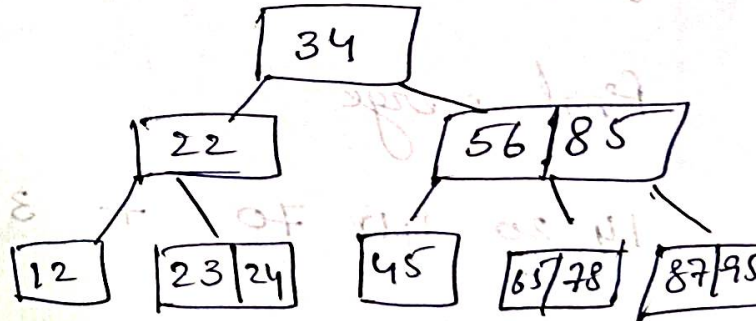
Insert 78



Insert 22:



final Tree



7) Merge Sort

40, 20, 70, 14, 60, 61, 97, 30

40, 20, 70, 14 | 60, 61, 97, 30

40 20 | 70 14 | 60 | 61 | 97 30

40 | 20 | 70 | 14 | 60 | 61 | 97 | 30

Merge :

40 + 20 → 20 40 60 + 61 → 60 61

70 + 14 → 14 70 97 + 30 → 30 97

20 40 | 14 70 | 60 61 | 30 97

Merge :

20 40 + 14 70 = 14 20 40 70

60 61 + 30 97 = 30 60 61 97

final merge

14 20 40 70 + 30 60 61 97

14 20 30 40 60 61 70 97

8) Quick Sort.

78 23 31 88 43 55 67 55

Pivot = 78

23 31 43 55 67 55 78 88

Step 2 : sort left

23 31 43 55 67 55

23 31 43 55 55 67

final answer

23 31 43 55 55 67 78 88

9) Open hashing & Linear Probing

0, 1, 81, 4, 64, 25, 16, 36, 9, 49

hash function:

$$h(k) = k \bmod 10$$

0 → 0

1 → 1 → 81

4 → 4 → 64

5 → 25

6 → 16 → 36

9 → 9 → 49

Linear Probing

Index	value
0	0
1	1
2	81
3	49
4	4
5	64
6	25
7	16
8	36
9	9

10) BFS

from diagram :

0 - 1
 0 - 3
 0 - 2
 3 - 4
 2 - 5
 4 - 5

steps:

start: 0

visit: 1, 3, 2

Then: 4, 5

final BFS Traversal: (4, 5)

0 → 1 → 3 → 2 → 4 → 5

BFS Tree

